

1. NAVIGATION & ORIENTATION

```
~/Developer/  
├── Statistics/  
├── Survival_Walls/  
├── Machine_Learning/  
└── Pandas_Practice/
```

PWD (PRINT WORKING DIRECTORY)

```
pwd
```

Shows your exact current location.

LS (LIST)

```
ls -la
```

Lists ALL files (even hidden .git) + permissions.

```
ls -lh
```

Lists file sizes in KB/MB/GB.

CD (CHANGE DIRECTORY)

```
cd ~/Developer/Machine_Learning
```

Moves you into the specified folder.

```
cd ..
```

Moves exactly ONE folder up.

2. FILES, FOLDERS & ARCHIVES

MKDIR / TOUCH

```
mkdir new_project
```

Creates a new folder.

```
touch config.yaml
```

Creates an empty file instantly.

CP / MV

```
cp data.csv backup.csv
```

Duplicates a file.

```
mv data.csv ./Clean_Data/
```

Moves a file (or renames it).

ZIP / UNZIP

```
zip -r archive.zip ./my_folder/
```

Compresses a whole folder recursively.

```
unzip dataset.zip
```

Extracts the archive here.

RM -R (REMOVE RECURSIVE)

```
rm -r Old_Project_Folder
```

Deletes folder & EVERYTHING inside FOREVER.

⚠️ DANGER: Bypasses Trash! Check `pwd` first.

3. DATA EXPLORATION (KILLER CMDS)

HEAD (TOP ROWS)

```
head -n 5 sales_data.csv
```

Prints the first 5 rows of a file to the screen.

Use when: Checking CSV headers without opening Excel.

TAIL (BOTTOM ROWS / LIVE)

```
tail -n 10 app.log
```

Prints the last 10 rows.

```
tail -f training.log
```

Follows the file live as new lines are added!

CAT (CONCATENATE)

```
cat .gitignore
```

Dumps the entire file content into the terminal.

Use when: Quickly reading small config files.

LESS (READ LARGE FILES)

```
less massive_dataset.json
```

Opens huge files in a scrollable viewer without crashing the terminal. (Press 'q' to quit).

4. SEARCH & DISCOVERY

FIND

```
find . -name "*.csv"
```

Searches current folder (.) for all CSV files.

GREP (SEARCH INSIDE FILES)

```
grep "import pandas" *.py
```

Finds exact text string inside all Python files.

5. PYTHON ENVIRONMENTS

VIRTUAL ENVIRONMENT

```
python -m venv venv
```

Creates isolated sandbox.

```
source venv/bin/activate
```

Turns sandbox ON (Prompt shows `(venv)`).

PIP

```
pip install -r requirements.txt
```

Installs bulk packages from list.

```
pip freeze > requirements.txt
```

Saves your current environment state.

6. HOMEBREW (MAC PACKAGE MGR)

BREW INSTALL / SEARCH

```
brew install git python
```

```
brew search postgresql
```

Finds and installs system tools.

BREW UPDATE & UPGRADE

```
brew update && brew upgrade
```

Updates Homebrew itself, then upgrades all tools.

BREW DOCTOR

```
brew doctor
```

Diagnoses system issues if packages refuse to work.

Use when: "Command not found" after installing.

7. DS LAUNCH & SYSTEM

IDE LAUNCHERS

```
code .
```

Opens current folder in VS Code.

```
jupyter lab
```

Launches Jupyter Lab in browser.

```
open -a RStudio
```

Launches RStudio application.

PROCESS CONTROL

```
top
```

Live task manager (press 'q' to quit).

```
kill -9 1234
```

Force kills frozen process by its PID number.

8. NETWORK & SSH

PING / CURL

```
ping github.com
```

Checks internet connection stability (CTRL+C to stop).

```
curl ifconfig.me
```

Returns your public IP address.

SSH GITHUB AUTH

```
ssh-keygen -t ed25519 -C "mail@.com"
```

Generates a new secure SSH key for GitHub access.

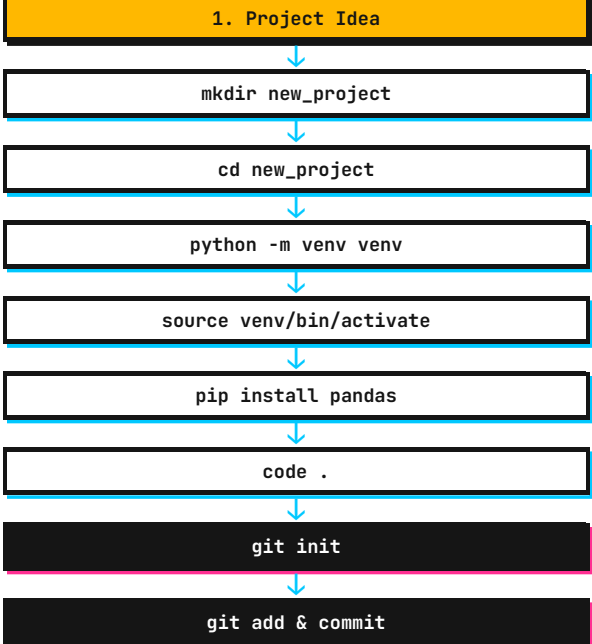
```
ssh -T git@github.com
```

Tests if your GitHub SSH connection is working.

9. SUPERPOWERS

- TAB** Autocomplete. Never type full names.
- CTRL+C** Kill Process. Stops a running script.
- CTRL+L** Clear Screen. Pushes clutter out of view.
- CTRL+R** Reverse Search. Finds past commands.
- ↑** Cycle History. Arrow up brings back commands.

10. CORE DS WORKFLOW



11. EMERGENCY ZONE

WHAT TO DO WHEN IT'S BROKEN

1. "WHAT DID I JUST BREAK?"

```
git status
```

Tells you which files changed or were deleted.

2. "WHERE THE HELL AM I?"

```
pwd
```

Ensure you didn't `cd` into system folders.

3. "WHAT DID I JUST TYPE?!"

```
history | tail -n 10
```

Shows your last 10 executed commands.

12. MANTRAS

CHECK WHERE YOU ARE (PWD)
BEFORE YOU DELETE (RM -R)!

THE TERMINAL DOES EXACTLY
WHAT YOU TELL IT TO DO.
NOT WHAT YOU MEANT.